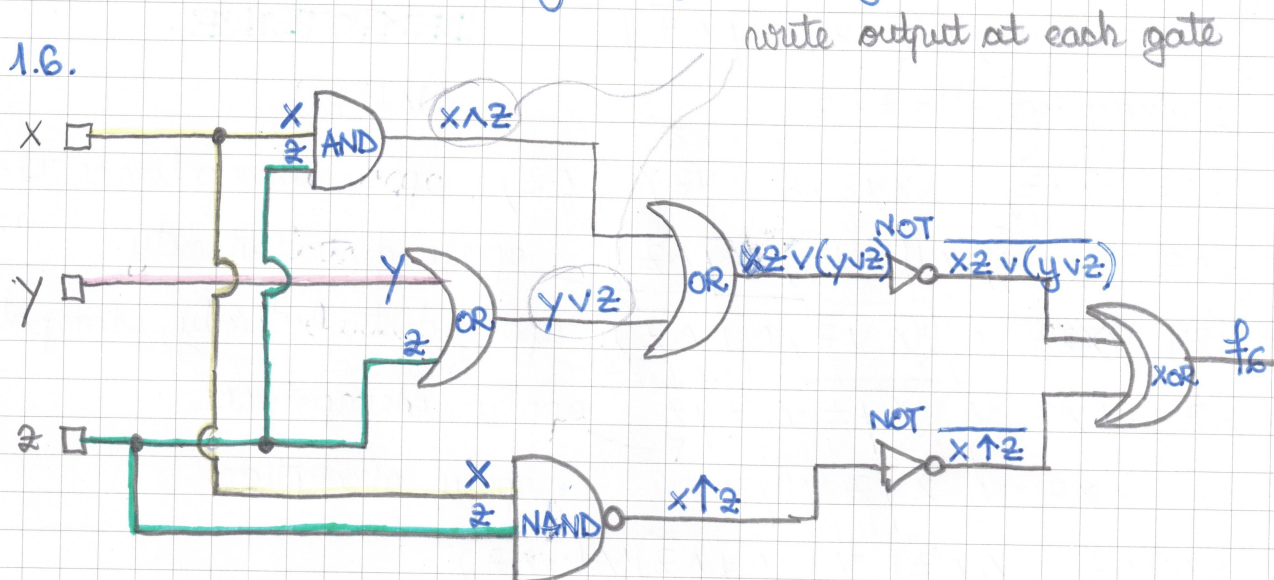


(Ex 1) Write the corresponding Boolean fn associated to the following logic circuit, then simplify it and draw simple equivalent circuit using only basic gates.

1.6.



$$f_6(x, y, z) = \overline{(xz \vee (y \vee z))} \oplus \overline{x \uparrow z}$$

replace $\oplus$

$$= (\overline{xz \vee (y \vee z)}) (\overline{x \uparrow z}) \vee (xz \vee (y \vee z)) (x \uparrow z)$$

$$= (xz \vee y \vee z) (\overline{x \uparrow z}) \vee \overline{xz \vee y \vee z} (x \uparrow z)$$

replace $\uparrow$

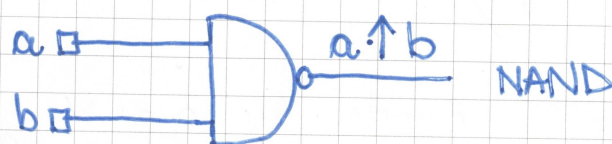
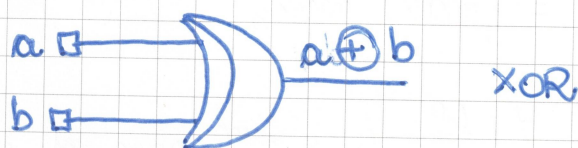
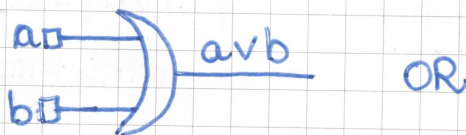
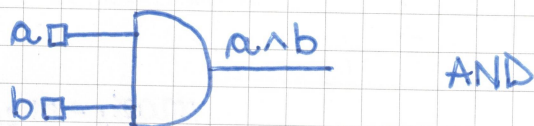
$$= (xz \vee y \vee z) (\overline{x \uparrow z}) \vee \overline{(xz \vee y \vee z)} (\overline{x \vee z})$$

$$= (xz \vee y \vee z) (x \wedge z) \vee \overline{((\overline{x \vee z}) \wedge \overline{y} \wedge \overline{z})} (\overline{x \vee z})$$

$$= (xz \vee y \vee z) xz \vee \overline{((\overline{x \vee z}) \wedge \overline{y} \wedge \overline{z})} (\overline{x \vee z})$$

GATES

identify gates given



$$a \oplus b = \overline{a}b \vee a\overline{b}$$

$$a \uparrow b = \overline{a \wedge b} = \overline{a} \vee \overline{b}$$

write formulas you use

$$= xz(xz \vee y \vee \bar{z}) \vee ((\bar{x} \vee \bar{z}) \wedge \bar{y} \wedge \bar{z})(\bar{x} \vee \bar{z}) \quad \text{ABSORPTION}$$

$$= (x \wedge z)((x \wedge z) \vee y \vee \bar{z}) \vee (\bar{x} \vee \bar{z})(\bar{x} \vee \bar{z}) \wedge \bar{y} \wedge \bar{z} \quad U \wedge (U \vee V) \equiv U$$

$$= (x \wedge z) \wedge ((x \wedge z) \vee y \vee \bar{z}) \vee ((\bar{x} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge \bar{y} \wedge \bar{z}) \quad U \vee (U \wedge V) \equiv U$$

$$\vee ((\bar{x} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge \bar{y} \wedge \bar{z}) \quad \text{IDEMPOTENCE}$$

apply absorption $U = z$

$$U \wedge U \equiv U$$

$$= (x \wedge z)(y \vee \bar{z}) \vee ((\bar{x} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge \bar{y} \wedge \bar{z}) \quad \text{apply idempotence } U = \bar{x} \vee \bar{z}$$

$$= (x \wedge z \wedge (y \vee \bar{z})) \vee ((\bar{x} \vee \bar{z}) \wedge \bar{y} \wedge \bar{z}) \quad \text{apply distributivity}$$

$$= xzy \vee xz\bar{z} \vee ((\bar{x} \vee \bar{z}) \wedge \bar{y} \wedge \bar{z}) \quad \text{apply distributivity, idempotence}$$

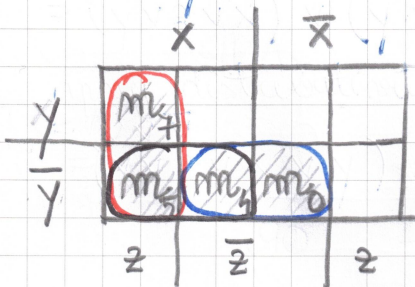
$$= xzy \vee xz \vee \bar{x}y\bar{z} \vee \bar{z}\bar{y}\bar{z} \quad \text{apply idempotence}$$

$$* = xzy \vee xz \vee \bar{x}y\bar{z} \vee \bar{y}\bar{z} \quad \text{apply absorption}$$

$$= (x \wedge y \wedge z) \vee (x \wedge z) \vee (\bar{x} \wedge \bar{y} \wedge \bar{z}) \vee (\bar{y} \wedge \bar{z})$$

$$= (x \wedge z) \vee (\bar{y} \wedge \bar{z})$$

$$= xz \vee \bar{y}\bar{z} \quad \leftarrow \text{we must check that it is the simplest form}$$



$$\max_1 = m_4 \vee m_0 = \bar{y}\bar{z}$$

$$\max_2 = m_4 \vee m_5 = x\bar{y}$$

$$\max_3 = m_5 \vee m_7 = xz$$

$$M(f) = \{\max_1, \max_2, \max_3\}$$

$$C(f) = \{\max_1, \max_3\}$$

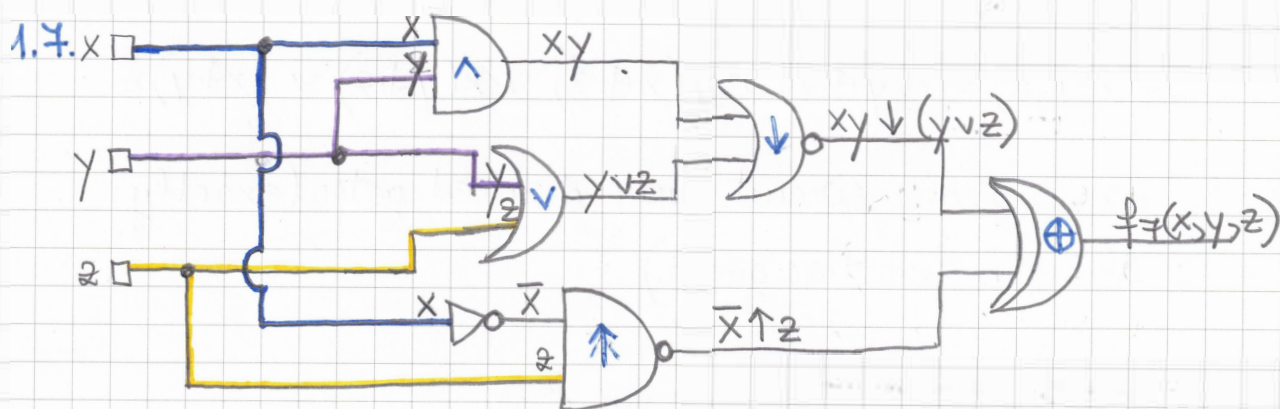
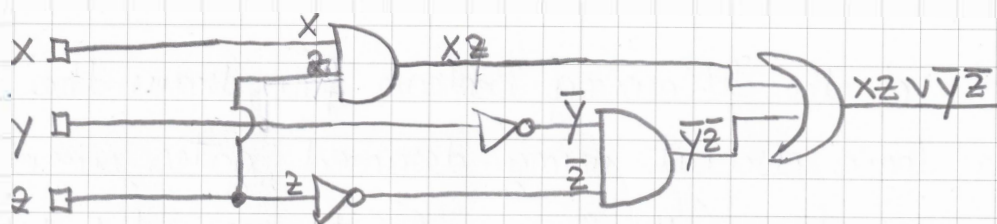
$$\left. \begin{array}{l} M(f) \neq C(f) \\ C(f) \neq \emptyset \end{array} \right\} \Rightarrow 2^{\text{nd}} \text{ case of simplification}$$

$\Rightarrow$ We denote by $g = \bigvee_{\max \in C(f)} \max$ the disjunction

of central monoms belonging to every simplified form

$g = \max_1 \vee \max_3 \Rightarrow \max_1, \max_3$ cancel all minterms $\Rightarrow$

$$\Rightarrow f_6^S(x, y, z) = \max_1 \vee \max_3 \\ = xz \vee \bar{y}\bar{z}$$



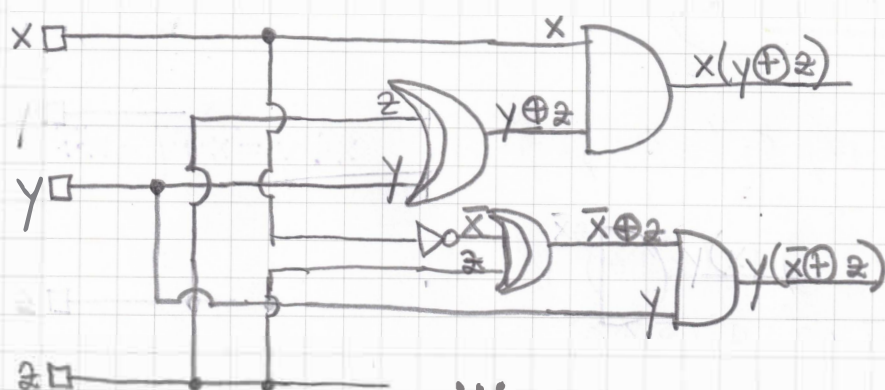
$$f_7(x, y, z) = (xy \downarrow (y \vee z)) \oplus (\bar{x} \uparrow z)$$

$$x \downarrow y = \overline{x \vee y} = \bar{x} \bar{y}$$

Ex 2 For each of the following Boolean fms, draw the corresponding logic circuit using derived gates, simplify the function and draw the logic circuits associated to all simplified forms of the initial fm using basic gates.

2.7. $f_7(x, y, z) = x(y \oplus z) \vee y(\bar{x} \oplus z) \vee x(y \downarrow z) \vee (x \downarrow y)\bar{z}$

Step 1: draw logic circuit using derived gates (exactly like the function given)



draw circuit for $x(y \downarrow z)$

$(x \downarrow y)\bar{z}$

and then $\Rightarrow f_7$

Step 2: simplify fm

$$\begin{aligned}
 f(x, y, z) &= x(y\bar{z} \vee \bar{y}z) \vee y(xz \vee \bar{x}\bar{z}) \vee x(\bar{y} \wedge \bar{z}) \vee (\bar{x} \wedge \bar{y})\bar{z} \\
 &= \underset{m_6}{x\bar{y}\bar{z}} \vee \underset{m_2}{x\bar{y}z} \vee \underset{m_7}{xy\bar{z}} \vee \underset{m_5}{x\bar{y}z} \vee \underset{m_4}{x\bar{y}\bar{z}} \vee \underset{m_0}{x\bar{y}\bar{z}}
 \end{aligned}$$

x \ yz	00	01	11	10
0	m ₀			m ₂
1	m ₄	m ₅	m ₇	m ₆

max₁ (m₀, m₂, m₄, m₆)
max₂ (m₀, m₄)
max₃ (m₂, m₆)

x \ yz	00	01	11	10
0	m ₀			m ₂
1	m ₄	m ₅	m ₇	m ₆

not okay, can double factorize

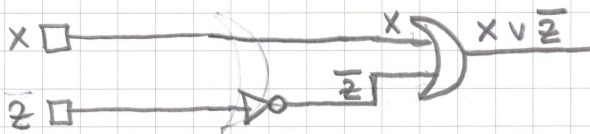
$$\max_1 = m_0 \vee m_4 \vee m_2 \vee m_6 = \bar{z}$$

$$\max_2 = m_4 \vee m_5 \vee m_7 \vee m_6 = x$$

$$M(f) = C(f) = \{ \max_1, \max_2 \} \Rightarrow 1^{\text{st}} \text{ case of simplification}$$

$$\Rightarrow f_7^5(x, y, z) = \max_1 \vee \max_2$$

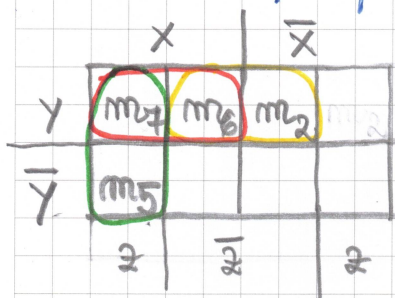
$$= x \vee \bar{z}$$



$$f_7(x, y, z) = (xy \downarrow (y \vee z)) \oplus (\bar{x} \uparrow z)$$

$$= (\bar{y} \wedge \bar{z}) \oplus (x \vee \bar{z})$$

$$= xy \vee y\bar{z} \vee x\bar{z}$$



$$\begin{array}{l} xy: \begin{array}{ccc} 1 & 1 & 1 \\ x & y & z \end{array} \\ y\bar{z}: \begin{array}{ccc} 1 & 1 & 0 \\ y & \bar{z} & \end{array} \\ x\bar{z}: \begin{array}{ccc} 1 & 0 & 1 \\ x & \bar{z} & \end{array} \end{array}$$

$$\text{max}_1 = m_6 \vee m_2 = y\bar{z}$$

$$\text{max}_2 = m_6 \vee m_7 = xy$$

$$\text{max}_3 = m_7 \vee m_5 = x\bar{z}$$

$$M(f) = \{\text{max}_1, \text{max}_2, \text{max}_3\}$$

$$C(f) = \{\text{max}_1, \text{max}_3\}$$

$$\left. \begin{array}{l} M(f) \neq C(f) \\ C(f) \neq \emptyset \end{array} \right\} \Rightarrow 2^{\text{nd}} \text{ case of simplification}$$

$$g = \bigvee_{m \in C(f)} m = \text{max}_1 \vee \text{max}_3 = y\bar{z} \vee x\bar{z}$$

→ all minterms covered ⇒ 1 unique simplified form

$$f_7^s(x, y, z) = y\bar{z} \vee x\bar{z}$$

$$a \downarrow b = \overline{a \vee b} = \bar{a} \bar{b}$$

$$xy \downarrow (y \vee z) =$$

$$= \overline{xy \vee y \vee z}$$

$$= (\bar{x} \vee \bar{y}) \wedge \bar{y} \wedge \bar{z}$$

$$= \bar{y} \wedge \bar{z}$$

$$\bar{x} \uparrow z = \overline{\bar{x} \wedge z} = x \vee \bar{z}$$

$$a \uparrow b = \overline{a \wedge b} = \bar{a} \vee \bar{b}$$

$$a \oplus b = \bar{a}b \vee a\bar{b}$$

$$\bar{y}\bar{z} \oplus (x \vee \bar{z}) =$$

$$= \overline{\bar{y}\bar{z}} \wedge (x \vee \bar{z}) \vee \bar{y}\bar{z} \wedge \overline{x \vee \bar{z}}$$

$$= ((y \vee z) \wedge (x \vee \bar{z})) \vee \underbrace{\bar{y}\bar{z} \wedge x}_{0}$$

$$= (y \vee z) \wedge (x \vee \bar{z})$$

$$= xy \vee y\bar{z} \vee xz \vee \underbrace{z\bar{z}}_0$$

$$= xy \vee y\bar{z} \vee xz$$